

EXTREME VALUES, QUANTILES, AND VALUE AT RISK

MEASURING THE RISK IN THE TAIL OF FINANCIAL RETURNS

Financial Econometrics

Week 3

WHERE WE ARE

The whole course has repeated one empirical fact:

The recurring finding

Financial returns are **fat-tailed**. Extreme losses happen far more often, and are far larger, than a normal distribution predicts.

Today we turn that fact into a **risk number**.

- Define **Value at Risk (VaR)** and **Expected Shortfall (ES)**.
- Estimate them **four ways**: RiskMetrics, GARCH, empirical quantiles, EVT.
- See *how badly* normal-based numbers understate deep-tail danger.
- Correct EVT for the fact that extremes **cluster** (the extremal index).

OUTLINE

- ➊ Value at Risk and Expected Shortfall
- ➋ RiskMetrics
- ➌ An Econometric (GARCH) Approach
- ➍ Quantile Estimation
- ➎ Extreme Value Theory
- ➏ Peaks Over Threshold
- ➐ When Extremes Cluster: the Extremal Index
- ➑ Synthesis

WHAT IS VALUE AT RISK?

Definition — Value at Risk

The **VaR** at level p over horizon h is the loss that will be exceeded with probability p . A 1-day 99% VaR of \$1m means: on 99 days out of 100 the loss stays under \$1m; on the 100th it may be worse.

Formally, if L is the loss over the horizon,

$$P(L > \text{VaR}_p) = p,$$

so VaR_p is the upper p -quantile of the loss distribution.

Three ingredients:

- 1 the **horizon** (1 day, 10 days);
- 2 the **confidence** ($p = 1\%$ or 5%);
- 3 an estimate of the **loss distribution**.

Every method in this lecture differs only in that last ingredient.

THE SIMPLEST ESTIMATES: HISTORICAL VS. NORMAL

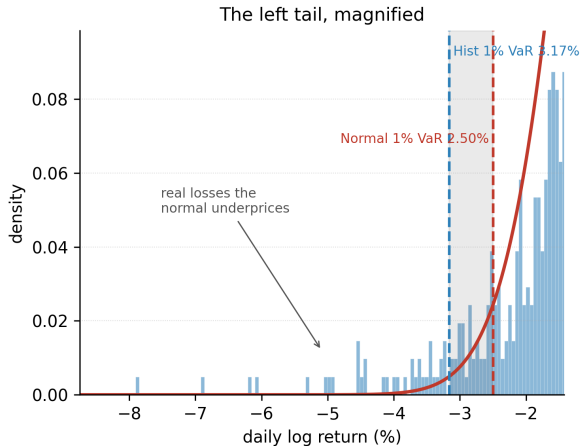
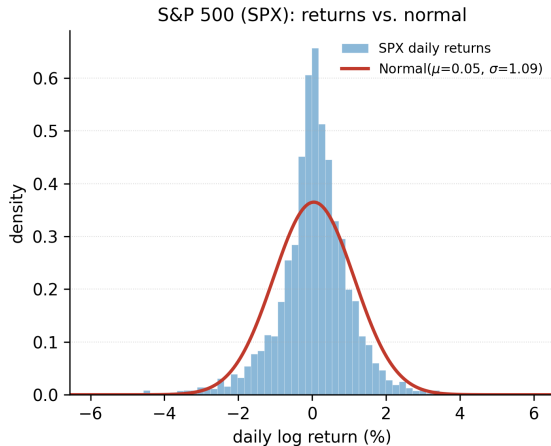
Historical VaR = the empirical quantile of past losses. **Normal VaR** = $-(\mu + z_p \sigma)$.

Ticker	Hist 5%	Norm 5%	Hist 1%	Norm 1%	Hist ES 1%	Norm ES 1%
AAPL	2.71	2.85	4.92	4.07	6.80	4.68
MSFT	2.54	2.64	4.28	3.77	6.19	4.33
AMZN	3.11	3.32	5.69	4.73	7.82	5.43
SPX	1.65	1.75	3.17	2.50	4.63	2.87
GLD	1.67	1.72	3.03	2.44	4.26	2.80

Table: 1-day VaR and ES, historical vs. normal (% loss).

At 5% the two agree. At 1% the normal falls short — and the deeper we look, the worse it gets. That is the whole problem, in one table.

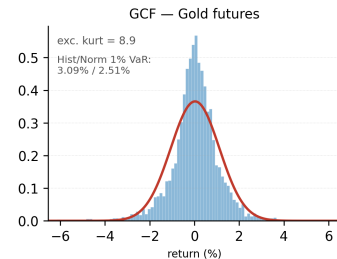
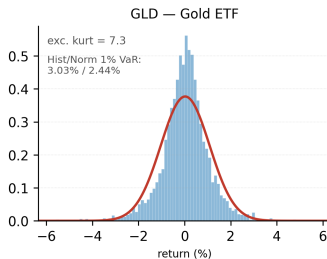
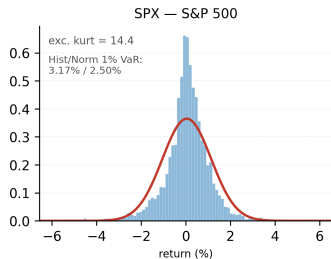
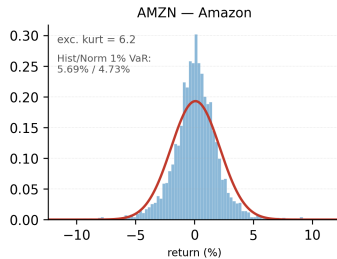
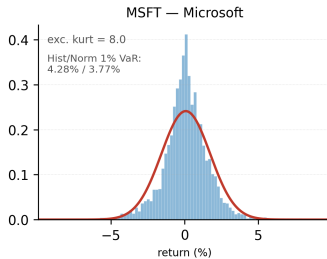
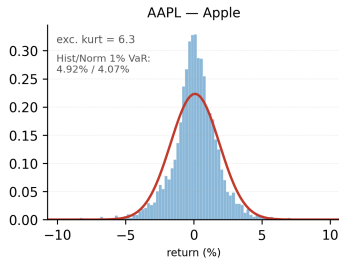
SEEING THE TAIL PROBLEM



Same σ , wrong *shape*: the normal has a taller, thinner peak and **far too little mass in the tail**. Real loss days pile up beyond where its density has effectively vanished — so its 1% VaR (2.50%) sits *inside* the true 1% loss (3.17%).

EVERY SERIES TELLS THE SAME STORY

Every series: fat peak, fat tails — the normal (red) misses both



All six are **leptokurtic** (excess kurtosis 6–14): a taller peak and heavier tails than the normal. The historical 1% VaR

VaR'S BLIND SPOT $\Rightarrow$ EXPECTED SHORTFALL

VaR is a **threshold**: it says nothing about *how bad* losses beyond it are.

Definition — Expected Shortfall (ES)

The **ES** (or conditional VaR) is the *average* loss in the worst $p\%$ of cases:

$$\text{ES}_p = E[L \mid L > \text{VaR}_p].$$

- Captures tail **severity**, not just frequency.
- Is **coherent**: a diversified book never has higher ES than the sum of parts (VaR is *not* coherent).
- Under conditional normality: $\text{ES}_p = -\mu + \sigma \frac{\phi(z_p)}{p}$.

Because ES lives *further* out, the normal underestimates it even more (S&P: 4.63% vs. 2.87%).
Regulators have moved from VaR toward ES for exactly this reason.

A MAP OF THE FOUR METHODS

Historical and normal share a deeper flaw beyond thin tails: both are **static** — one unconditional number, ignoring that volatility *clusters*.

Each method below repairs a specific flaw in the previous one:

- ① **RiskMetrics** — make volatility time-varying (but keep normality).
- ② **Econometric / GARCH** — fix forced persistence *and* the thin tail.
- ③ **Quantile methods** — strip back to the essence: VaR *is* a quantile.
- ④ **Extreme Value Theory** — model the tail *shape*, where data run out.

We take them in that order. Each says what it fixes and what it leaves open.

RISKMETRICS: TIME-VARYING VOLATILITY, NORMAL TAIL

Definition — RiskMetrics

Model volatility with an **exponentially weighted moving average (EWMA)** and assume conditionally normal returns.

The volatility recursion:

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) a_{t-1}^2, \quad \lambda = 0.94 \text{ (daily).}$$

- Today's variance = decaying weighted average of past squared returns.
- Exactly a **GARCH(1,1)** with $\alpha_0 = 0$, $\alpha_1 = 1 - \lambda$, $\beta_1 = \lambda$ — an **IGARCH** with persistence $\alpha_1 + \beta_1 = 1$.
- VaR is a normal quantile of today's volatility: $\text{VaR}_p = -z_p \sigma_t$ (drift ignored short-horizon).

RISKMETRICS: STRENGTHS AND WEAKNESSES

Current 1-day 99% figures:

Ticker	EWMA σ_T	RiskMetrics 99% VaR
AAPL	2.11	4.92
SPX	0.95	2.20
GLD	1.73	4.02

Strength: adapts to current conditions — the S&P's low σ_T gives a tight 2.2% VaR because July 2026 was calm.

Two weaknesses (the ones this course keeps meeting):

- forces persistence to *exactly* 1;
- assumes **normality** — inherits the thin-tail error.

RISKMETRICS FOR A PORTFOLIO: DIVERSIFICATION

For weights w and covariance matrix Σ :

$$\sigma_p^2 = w^\top \Sigma w, \quad \text{VaR}_p = -z_p \sigma_p.$$

Because correlations are below one, **diversification lowers VaR**.

50/50 S&P–gold book

The two are almost **uncorrelated** ($\text{corr} = 0.06$), so the combined 99% VaR is **1.79%** — well below the **2.47%** you get by adding the two positions' VaRs. The 0.68-point gap is the diversification benefit.

This coherence property is shared by ES — but *not* by VaR in general.

FROM FIXED λ TO AN ESTIMATED MODEL

RiskMetrics is a fixed- λ special case. The **econometric approach** replaces it with a properly estimated volatility model and a properly chosen innovation:

$$\text{VaR}_t(p) = -(\mu + q_p \sigma_t),$$

- $\sigma_t = \mathbf{GARCH}$ conditional volatility (fixes forced persistence);
- $q_p =$ quantile of the innovation — normal, or the fatter **Student- t** (fixes the thin tail).

What we already saw

Normal-innovation VaR was breached far too often; **t -innovations restored honest backtest coverage**. This is the recommended everyday VaR.

MULTI-PERIOD VaR: THE SQUARE-ROOT-OF-TIME RULE

If returns are iid, variance adds, so over h days

$$\text{VaR}_p(h) \approx \sqrt{h} \sigma (-q_p) - h\mu.$$

The S&P's 10-day 99% VaR $\approx \sqrt{10} \approx 3.2$ times the one-day figure.

Caveat under GARCH

The h -step variance forecast **mean-reverts**. So:

- from an *elevated* start, true VaR grows *slower* than $\sqrt{h}$;
- from a *calm* start, *faster*.

$\sqrt{h}$ is a convenient approximation, not a law.

CONDITIONAL EXPECTED SHORTFALL

Combine the normal ES formula with a GARCH volatility:

$$\text{ES}_t(p) = -\mu + \sigma_t \frac{\phi(z_p)}{p}.$$

- On a **calm day** the shortfall is small.
- After a **shock** it widens.

Same adaptivity as conditional VaR, now measuring the *severity* of tail events rather than their threshold.

STEP BACK: VaR IS JUST A QUANTILE

The methods so far are increasingly elaborate *parametric* machines. Underneath them all sits one object:

Definition — the p -quantile

The value below which a fraction p of outcomes fall. **VaR is a quantile of the loss distribution — so estimating VaR is estimating a quantile.**

Two direct, low-assumption answers:

- **unconditional** — order statistics (the empirical quantile);
- **conditional** — quantile regression.

ORDER STATISTICS = HISTORICAL VaR

Sort the returns $r_{(1)} \leq r_{(2)} \leq \dots \leq r_{(n)}$. The empirical p -quantile is about $r_{(\lceil np \rceil)}$.

- **Strength:** no distributional assumption.
- **Weakness — the tail:** the 1% quantile of 3,770 days rests on ~ 38 observations; the 0.1% quantile on *fewer than four*.

The key limitation

Historical VaR is noisy exactly where it matters, and **cannot exceed the worst loss ever observed**. That limitation is the entire motivation for Extreme Value Theory.

QUANTILE REGRESSION

Definition — quantile regression (Koenker–Bassett)

Models a conditional *quantile* of the response, where ordinary regression models the conditional *mean*.

Minimise the asymmetric **check-function** loss:

$$\min_{\beta} \sum_t \rho_{\tau}(r_t - x_t^{\top} \beta), \quad \rho_{\tau}(u) = u (\tau - \mathbf{1}\{u < 0\}).$$

Set $\tau = 0.01$ and regress the return quantile on covariates (lagged volatility, a volatility index): a VaR that **moves with the market without assuming a distribution**. A semiparametric middle ground between GARCH VaR and historical VaR.

WHERE THE DATA RUN OUT

Order statistics gave historical VaR — silent beyond the worst day seen. Quantile regression still needs data wherever we look.

Definition — Extreme Value Theory (EVT)

Describes the distribution of the *most extreme* observations, giving a principled way to model — and **extrapolate into** — the far tail.

Instead of *counting* tail observations it never has, EVT models the tail's **shape** and projects it outward.

BLOCK MAXIMA AND THE GEV DISTRIBUTION

Take the maximum of each block of n iid observations. The **Fisher–Tippett–Gnedenko theorem**: suitably rescaled, this maximum can only converge to the **Generalized Extreme Value (GEV)** distribution:

$$G(x) = \exp\left[-\left(1 + \xi \frac{x-\mu}{\sigma}\right)^{-1/\xi}\right].$$

The **shape parameter** ξ sorts *all* tails into three families:

$\xi = 0$	Gumbel	thin tails (like the normal)
$\xi > 0$	Fréchet	heavy, power-law tails
$\xi < 0$	Weibull	bounded

The single number ξ decides how fat the tail is.

FINANCIAL RETURNS ARE FRÉCHET

Fitting the GEV: split losses into blocks (say monthly), take each block maximum, estimate (μ, σ, ξ) by maximum likelihood. The sign of $\hat{\xi}$ classifies the tail.

For financial losses the answer is always the same

$\hat{\xi} > 0$ — the **Fréchet** case. Heavy, power-law tails: exactly the fat tails of the distribution chapter, seen through the lens of extremes. Every one of our six series comes out Fréchet.

We will quantify $\hat{\xi}$ precisely with the more efficient peaks-over-threshold method shortly.

AN EXTREME-VALUE APPROACH TO VaR: EXTRAPOLATION

The point of EVT for risk is **extrapolation**.

- Historical VaR cannot exceed the worst observed loss.
- The normal has the wrong tail *shape*.
- A fitted extreme-value tail gives a sensible VaR at 0.1% or 0.01% — probabilities so small few or no observations reach them.

EVT models the *shape* of the tail from the moderately-extreme data and projects it outward.

Multi-period scaling: for heavy ($\xi > 0$) tails the h -period VaR grows roughly like h^ξ times the one-period tail — *slower* than the number of periods, and governed by the same shape parameter. A cleaner rule than the normal's $\sqrt{h}$.

RETURN LEVELS

Definition — N -period return level

The loss expected to be exceeded on average *once every N periods* — i.e. VaR at $p = 1/N$. A “10-year return level” is the daily loss seen about once a decade.

Computed from the S&P’s fitted tail:

Horizon	Daily loss
Once a year	4.4%
Once a decade	8.5%
Once every 40 years	12.2%

That last figure lands right on the actual worst day in the sample (−12.8%) — a reassuring calibration check.

A MODERN APPROACH: PEAKS OVER THRESHOLD (POT)

Block maxima waste data — one number per block.

Definition — peaks over threshold

Model *every* loss above a high threshold u . The **Pickands–Balkema–de Haan theorem**: these exceedances follow a **Generalized Pareto Distribution (GPD)**.

For a high enough u , the excess $Y = X - u$ (given $X > u$) has

$$P(Y \leq y \mid X > u) = 1 - \left(1 + \xi \frac{y}{\beta}\right)^{-1/\xi},$$

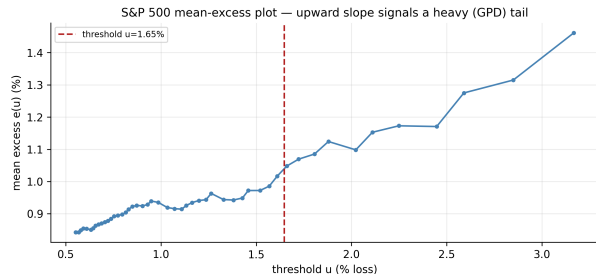
with the **same shape** ξ as the GEV and a **scale** β . **Two parameters describe the whole tail.**

CHOOSING THE THRESHOLD: THE MEAN-EXCESS FUNCTION

Definition — mean-excess function

$e(u) = E[X - u \mid X > u]$, the average overshoot above u . For a GPD tail it is **linear**:
 $e(u) = (\beta + \xi u)/(1 - \xi)$.

Practical rule: plot empirical mean excess vs. u ; pick the point beyond which it is a straight line.



The S&P plot straightens above roughly **1.5%**, so we set the threshold at the **95th percentile** of losses. Above it, the mean overshoot rises linearly — the signature of a heavy (GPD) tail.

FITTING THE GPD

S&P exceedances above $u = 1.65\%$ (189 of them), by maximum likelihood:

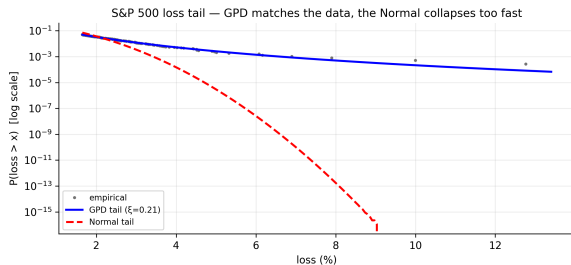
$$\hat{\xi} = 0.21, \quad \hat{\beta} = 0.81\%.$$

- $\hat{\xi} > 0$: the Fréchet, power-law tail.
- **Tail index** $1/\hat{\xi} \approx 4.7$ = effective number of finite moments.
- **Matches the Student- t d.o.f. $\hat{\nu} \approx 5.6$ estimated earlier** — two routes to the same tail heaviness.

Ticker	$\hat{\xi}$	$1/\hat{\xi}$	Ticker	$\hat{\xi}$	$1/\hat{\xi}$
AAPL	0.167	6.0	SPX	0.213	4.7
MSFT	0.203	4.9	GLD	0.251	4.0
AMZN	0.162	6.2	GCF	0.243	4.1

Every $\hat{\xi} > 0$; **gold has the heaviest tail** — “safe haven” means low *average* volatility, not a thin tail.

THE TAIL FIT, AND VaR FROM POT



Inverting the GPD tail gives closed-form extreme VaR and ES:

$$\text{VaR}_p = u + \frac{\beta}{\xi} \left[\left(\frac{n}{N_u} p \right)^{-\xi} - 1 \right],$$

$$\text{ES}_p = \frac{\text{VaR}_p}{1 - \xi} + \frac{\beta - \xi u}{1 - \xi}.$$

The empirical exceedances sit on the GPD line; the normal tail collapses far too fast, understating extreme losses by *orders of magnitude*.

THE PUNCHLINE: DEEP-TAIL COMPARISON

S&P 500 VaR by method:

Level p	EVT (GPD)	Normal	Historical
1.0%	3.21	2.50	3.17
0.5%	4.06	2.77	3.96
0.1%	6.61	3.33	6.33

At $p = 0.1\%$

EVT VaR is **6.6%** against the normal's **3.3%** — the normal understates the once-in-1000-day loss by half.

EVT agrees with historical where data exist, but **extrapolates** smoothly beyond any observed loss (its 99.9% ES is 9.0%).

THE BEST OF BOTH: GARCH–EVT (FILTERED EVT)

Plain POT misses **time variation** — threshold and scale are constant, but volatility clusters. The state of the art, in three steps:

- ① Fit a **GARCH** to strip out volatility clustering.
- ② Apply **POT–GPD** to the standardised residuals (now iid).
- ③ **Rescale** the extreme quantile by today's GARCH volatility σ_t .

This marries the two halves of the course

GARCH for the time-varying volatility, **EVT** for the tail shape. Best-performing VaR method in practice.

EXTREMES ARE NOT INDEPENDENT

Everything so far quietly assumed exceedances were **independent**. They are not.

Volatility clusters, and extreme losses cluster with it: the S&P's worst stretch was not one bad day but a **15-day run** in February–April 2020.

Definition — extremal index $\theta \in (0, 1]$

Measures how much a stationary series' extremes cluster:

- $\theta = 1$: extremes arrive independently;
- $\theta < 1$: they come in clusters, and $1/\theta$ is the **mean cluster size**.

THE $D(u_n)$ CONDITION

For the maximum of a *stationary* (not iid) series to still converge to a GEV law, its long-range dependence must fade at high levels — Leadbetter's **$D(u_n)$ condition** (a mixing condition). Under it,

$$P(M_n \leq u_n) \longrightarrow G(x)^\theta.$$

A stationary series' extremes behave like those of an iid series of the reduced effective length $n\theta$.

A stronger condition $D'(u_n)$ additionally forbids *local* clustering; when it holds, $\theta = 1$.

- Financial returns: $D(u_n)$ holds (returns mix) but $D'(u_n)$ *fails* (volatility clustering bunches extremes), so $\theta < 1$.

ESTIMATING THE EXTREMAL INDEX

- **Blocks method:** $\hat{\theta} \approx \frac{\# \text{ blocks with an exceedance}}{\text{total exceedances}}$.
- **Intervals (Ferro–Segers):** uses the *gaps* between exceedances — long gaps push $\theta \rightarrow 1$, short gaps (clusters) pull it down.

Ticker	$\hat{\theta}$	$1/\hat{\theta}$	Ticker	$\hat{\theta}$	$1/\hat{\theta}$
AAPL	0.59	1.7d	SPX	0.48	2.1d
MSFT	0.58	1.7d	GLD	0.39	2.6d
AMZN	0.59	1.7d	GCF	0.54	1.8d

Every $\hat{\theta} < 1$: extremes **cluster** in all six series (2–3 day episodes). The S&P’s 189 exceedances form only ~ 120 distinct episodes; its largest single cluster is the 15-day 2020 crash — **one event, not fifteen**. Gold clusters most.

VaR FOR A STATIONARY SERIES: WHAT CHANGES

The extremal index tells us what to adjust and what to leave alone.

Unaffected

The **one-day (marginal) VaR** — it is a quantile of the marginal loss distribution, and θ describes *timing*, not the marginal tail.

Changed

- **Return levels / multi-period VaR:** the rate of *distinct* extreme events is θ times the marginal rate, so use $n\theta$ in place of n . Ignoring θ overstates how often new episodes occur.
- **Drawdown risk:** a breach tends to be followed by more (cluster size 2–3 days), so cumulative loss over a cluster is far worse than iid implies. The real danger of a crash is the *run*, not the single day.

WHY GARCH AND EVT ARE COMPLEMENTARY

The clean practical fix is **GARCH–EVT** again:

- Fitting GARCH first removes the volatility clustering.
- Standardised residuals are then close to iid ($\theta \approx 1$).
- Ordinary POT–EVT applies to them directly; rescaling by σ_t puts the clustering back in the *conditional* forecast.

The bridge

GARCH captures *when* volatility is high; **EVT** captures *how heavy* the tail is; the **extremal index** explains why you need both.

THE FOUR METHODS, SIDE BY SIDE

- **Historical** — model-free and reliable at 1–5%, but blind beyond the worst observed loss and noisy in the far tail.
- **Normal / RiskMetrics** — adapts to current volatility, but its thin tail *systematically understates* extreme losses (half the true 0.1% VaR).
- **GARCH- t** — adapts to volatility *and* uses a fat-tailed innovation; well-calibrated VaR.
- **EVT / GARCH-EVT** — targets the tail *shape* directly, extrapolates safely, and combined with GARCH is the most accurate of all.

The message the course has built toward

The tail is fatter than the bell curve allows, and pretending otherwise is exactly the mistake that shows up as a blown risk limit on the worst day.

KEY TAKEAWAYS

- **VaR** = loss exceeded with probability p ; **ES** averages the losses beyond it and is the coherent, regulator-preferred measure.
- **RiskMetrics** = EWMA volatility + normality — simple, adaptive, thin-tailed; diversification lowers portfolio VaR (S&P–gold corr 0.06).
- **Econometric** = GARCH volatility + t quantile; scales roughly like $\sqrt{h}$.
- Historical VaR is a **quantile** from order statistics — model-free but tail-limited; **quantile regression** adds covariates.
- **EVT**: block maxima $\rightarrow$ GEV; modern POT $\rightarrow$ GPD. Tails all heavy ($\hat{\xi} = 0.16\text{--}0.25$).
- **EVT VaR** extrapolates: at 0.1% the S&P is 6.6% vs. normal 3.3%. **GARCH–EVT** is the most accurate.
- Real extremes **cluster**: extremal index $\theta \approx 0.4\text{--}0.6$ (cluster size 1.7–2.6 days). One-day VaR unchanged; **drawdowns** worse; GARCH filtering restores independence.

Questions?